Worksheet 4

Bivariate distributions and linear fitting – DETAILED SOLUTIONS

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Detailed solution – Exercise 1

The sums are

$$\sum x = 101, \quad \sum y = 198, \quad \sum x^2 = 1141, \quad \sum y^2 = 4464, \quad \sum xy = 2251.$$

Thus

$$\bar{x} = 10.1, \quad \bar{y} = 19.8,$$

$$V(x) = 1141/10 - 10.1^2 = \boxed{12.09}, \quad V(y) = 4464/10 - 19.8^2 = \boxed{54.36},$$

$$\sigma_x = 3.477, \quad \sigma_y = 7.373,$$

$$Cov(x, y) = 2251/10 - (10.1)(19.8) = \boxed{25.12},$$

$$R = 25.12/(3.477 \times 7.373) = \boxed{0.9799}.$$

Since $|R| \geq 0.86$, the course classifies this as a good positive linear correlation.

$$a = 25.12/12.09 = 2.07775, \quad b = 19.8 - 2.07775(10.1) = -1.18528,$$

so

$$\boxed{\hat{y} = 2.07775x - 1.18528}.$$

For 3000 customers, $x = 30$: $\hat{y} = 61.1472$, hence

$$\boxed{611\,472 \text{ euros approximately}}.$$

Detailed solution – Exercise 2

$\bar{t} = 5.5$, $V(t) = 8.25$, $\sum tx = 654$, so

$$Cov(t, x) = 654/10 - (5.5)(10.1) = 9.85,$$

$$R(t, x) = \boxed{0.9863}.$$

$$a = 9.85/8.25 = 1.19394, \quad b = 10.1 - 1.19394(5.5) = 3.53333.$$

Hence $\hat{x} = 1.19394t + 3.53333$. For $t = 12$, $\hat{x} = 17.8606$, about 1786 customers. Using the first regression, $\hat{y} = 35.9246$, or about $\boxed{359\,246 euros}$. **Detailed solution – Exercise 3**

$$A = ((1 + 2 + 3)/3, (7 + 10 + 10)/3) = (2, 9),$$

$$B = ((4 + 5)/2, (14 + 14)/2) = (4.5, 14).$$

The slope is $(14 - 9)/(4.5 - 2) = 2$ and the intercept is 5. Therefore the Mayer line is

$$\boxed{y = 2x + 5}.$$